



COMSATS University Islamabad, Lahore Campus
Department of Computer Engineering

Assignment 3 – FALL 2025

Course Title:	Microprocessor Systems and Interfacing				Course Code:	CPE 342	Credit Hours:	4(3,1)
Course Instructor:	Dr. Abbas Javed, Mr. Moazzam Ali Sahi				Program Name:	BCE		
Semester:	5 th	Batch:	FA23	Section:	A & B	Date Given:	December 12, 2025	
Submission Date:	December 15, 2025				Maximum Marks:	30		
Name:					Registration Number:			
<u>Important Instructions / Guidelines:</u>								
• Do your own work, PLAGARISM will be graded as ZERO. • No late submission.								

Question 1: [CLO2-PLO3-C5] [3+10+2 Marks]

Design a Stepper Motor Controller with Position Display and Time-Based Rotation using 8088-Microprocessor. Integrate a **stepper motor**, **7-segment display**, **two push buttons (A & B)**, and a **programmable timer** with the **8088 microprocessor** using **Programmable Peripheral Interface (PPI 8255)** to design a stepper motor control system.

Requirements:

- Button-Controlled Rotation:**
 - If **Button A** is pressed, the stepper motor should rotate **clockwise**.
 - If **Button B** is pressed, the stepper motor should rotate **anti-clockwise**.
- Position Display:**
 - Display the **current position** of the motor on a **7-segment display**.
 - The position should increment or decrement as the motor rotates.
- Timer-Based Automatic Rotation:**
 - Use a **programmable timer (e.g., 8254)** to rotate the motor automatically in the **clockwise direction every 1 second** if no button is pressed.
 - The timer should generate an interrupt to trigger the rotation.
- Integration:**
 - Implement the system using an **8088 microprocessor** with **PPI 8255** for interfacing the buttons, 7-segment display, and stepper motor.
 - Include **both manual and automatic rotation modes** without conflict.

Deliverables:

- Write an Assembly code for the 8088 microprocessor for interfacing push buttons, 7 segments, timers and stepper motor.
- Design a circuit diagram for the complete system. Also, explain your PPI, Timers configuration, and how you are displaying the position of the motor on the 7-segment display.